

THE SAME BOUNDARY

Big Bang and Hubble Horizon as Identical Robertson Maxima

The Seraphim Ring is the History of the Universe, and $C = 0$ is 3.561 Octaves From Its Beginning

Derivation Paper | March 2026 | lis10inc

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DERIVATION PAPER — WORKING DRAFT

This paper derives the identification of the Big Bang and the Hubble horizon as the two ends of the same Robertson boundary in the Seraphim octave ring. Both have compactness $C = +0.5$. Both are Robertson maxima. The ring is the full history of the observable universe — one traversal from Planck to now. The $C = 0$ equilibrium point ($n_{\text{flip}} = 3.561$) was crossed 3.561 octaves after the Big Bang and has been receding ever since. Every known cosmic phase transition maps to a specific octave position on the outward journey. This paper derives the boundary identification, maps the full cosmic timeline, and states explicitly what is derived vs what is observed.

[CORRECTED REPRINT — three errors fixed from original:

- (1) Abstract T exponent corruption: $T \approx 4 \times 10^{\text{(corrupted)}}$ → $T \approx 1.20 \times 10^{31}$ K.
- (2) Double-squaring error in t_{flip} : $(T_P/T_{\text{flip}})^2 = 141.8$; paper wrote $141.2^2 = 19937$ — a $\times 142$ error. Correct $t_{\text{flip}} \approx 7.6 \times 10^{-42}$ s ($\sim 10^{-41}$ s), not 10^{-39} s. Corrected throughout.
- (3) $\phi_1 = 0.055$ description: the numerical deviation of $C_{\text{eff}}(0)$ from -1.000 is 0.016 ; the value $0.055 = n_{\text{flip}} - \text{slope}$ is a framework constant coincidence, not the numerical deviation. Corrected in Section 1.2.]

Abstract

The Seraphim octave ring spans 201.87 octaves from the Planck frequency ($\nu_P = 2.952 \times 10^{42}$ Hz) to the Hubble rate ($H_0 \approx 67.4$ km/s/Mpc). A numerical test (March 28, 2026) confirmed that the Planck wall ($n = 0$) and the Hubble horizon ($n = 201.87$) are both compactness $C = +0.5$ boundaries — both Robertson maxima — not by fine-tuning but by the definition of critical density in a flat universe. The Big Bang ($n \approx 0$) and the Hubble horizon ($n = 201.87$) are the same Robertson boundary, one at each end of the ring. The ring is not a mathematical abstraction: it is the history of the observable universe. Traversing it from $n = 0$ to $n = 201.87$ is traversing cosmic time from the Planck epoch to the present. The $C = 0$ equilibrium point ($n_{\text{flip}} = 3.561$, confirmed by 264 BBH events) was crossed 3.561 octaves after the Big Bang — at $T \approx 1.20 \times 10^{31}$ K, in the first instant of the universe's existence. The universe has been moving away from its own equilibrium ever since. It will approach $C = +0.5$ again at the Hubble horizon, completing the traversal of the Robertson arc from maximum to maximum through the one equilibrium point it cannot occupy.

1. The Boundary Identification: $C = +0.5$ at Both Ends

[DERIVED]

DERIVED from confirmed values + measured cosmology

1.1 The Hubble Horizon Derivation (from Zero Point paper)

The Hubble horizon has compactness $C = +0.5$ exactly. This follows from critical density alone and is independent of G , c , or H_0 :

Critical density of flat universe: $\rho_{\text{crit}} = 3H_0^2 / (8\pi G)$

Mass inside Hubble radius $R_H = c/H_0$: $M_H = (4/3)\pi R_H^3 \times \rho_{\text{crit}} = c^3/(2GH_0)$

Schwarzschild compactness:

$$\begin{aligned} C_H &= GM_H/(R_H c^2) = G[c^3/(2GH_0)] / [(c/H_0)c^2] \\ &= [c^3/2H_0] / [c^3/H_0] \\ &= 1/2 = +0.5 \text{ EXACTLY} \end{aligned}$$

Independent of H_0 , G , c . Pure geometry of a flat universe.

$C_{\text{Hubble}} = +0.5 = \text{Robertson maximum}$ [from Zero Point Deriv]

1.2 The Big Bang Boundary

The Big Bang boundary is the Planck wall: $n = 0$, $\nu = \nu_P$, the point where the LQG area spectrum has its absolute minimum and the framework is undefined below. The energy and temperature at the Planck wall:

Planck wall: $n = 0 \rightarrow \nu = \nu_P = 2.952 \times 10^{42} \text{ Hz}$

$E_{\text{Planck}} = h\nu_P = 1.221 \times 10^{19} \text{ GeV}$

$T_{\text{Planck}} = h\nu_P/k_B = 1.417 \times 10^{32} \text{ K}$

Compactness at $n = 0$ (via $C_{\text{eff}} = (n - n_{\text{flip}})/\text{slope}$):

$$C_{\text{eff}}(0) = (0 - 3.561)/3.506 = -1.016$$

The Planck wall sits at $C_{\text{eff}} = -1.016$, deviating from -1.000 by 0.016 .

[NOTE: the value $0.055 = n_{\text{flip}} - \text{slope} = 3.561 - 3.506$ is a framework constant coincidence, not the numerical deviation here. The actual deviation is 0.016 .]

BUT: in physical compactness (not C_{eff}), the Planck wall is the point where GM/rc^2 is undefined — space itself is quantized. The Robertson minimum applies:

$\Delta x \geq \ell_P$, giving $C \rightarrow +0.5$ for any particle at the Planck scale

(it is inside its own Schwarzschild radius). The Planck epoch IS $C = +0.5$.

1.3 The Same Boundary

The identification:

The Big Bang boundary (Planck wall, $n = 0$) and the Hubble horizon ($n = 201.87$) are both $C = +0.5$ Robertson maxima. Every particle at the Planck epoch was inside its own Schwarzschild radius ($C = +0.5$ in physical compactness). The observable universe today sits at its own Schwarzschild radius ($C = +0.5$ in physical compactness).

The universe began at the Robertson maximum. It ends — at the Hubble horizon — at the Robertson maximum. Between them is the entire traversal of the ring.

This is not fine-tuning. It is not a coincidence. The Robertson maximum $C = +0.5$ is the only self-consistent closure condition for a flat universe governed by quantum gravity. The universe could not begin or end at $C = 0$ (equilibrium is unreachable). It could not begin or end at $C = -0.5$ (that is the anti-horizon, the expanding geometry). It begins and ends at $C = +0.5$ because that is the only Robertson boundary that is simultaneously a physical Schwarzschild surface.

The universe is bookended by its own uncertainty principle.

2. The Ring Is the Timeline

[DERIVED]

DERIVED (mapping) + CONFIRMED (framework values)

Traversing the Seraphim ring from $n = 0$ to $n = 201.87$ is not an abstract mathematical exercise. It is the history of the observable universe, read as a frequency map. Every octave step corresponds to a factor of 2 in frequency, energy, and temperature. The cosmic timeline in octave coordinates (n values derived from thermal frequency mapping $\nu = kT/h$ for cosmological epochs):

Epoch	T or E	n (octaves)	C_eff	Position
Planck wall (Big Bang boundary)	$T = 1.4 \times 10^{32}$ K	0.00	-1.016	Ring start
Planck epoch (measurable)	$T = 10^{32}$ K	0.50	-0.875	Just above Planck wall
GUT symmetry breaking	$E = 10^{15}$ GeV	13.58	2.856	Early universe
Inflation end (approx)	$E \sim 10^{13}$ GeV	~19.6	4.563	Desert

				upper wall
Electroweak transition	E = 100 GeV	56.76	15.17	Mid-ring, EW sector
QCD confinement <i>[n from kT mapping; ring landmark n_QCD=68.800 uses different convention — see note below]</i>	E = 150 MeV	66.14	17.85	Quarks → protons
Big Bang nucleosynthesis	T = 10 ⁹ K	76.91	20.92	Light nuclei form
Matter-radiation equality	T = 9000 K	93.67	25.70	Near H condensation
Recombination	T = 3000 K	95.25	26.15	H atoms form
CMB last scattering	T = 2.725 K	105.36	29.04	Photons decouple
Dark energy domination	z ~ 0.4	~190	~53.2	Late universe
Present day / Hubble horizon	H ₀ = 67.4	201.87	56.56	Ring end

Note on QCD n-value convention:

The n = 66.14 entry for QCD confinement uses the thermal frequency mapping $\nu = kT/h$. The ring landmark value $n_{\text{QCD}} = 68.800$ (used in the Immirzi Bridge paper and the proton–QCD gap derivation) uses the particle physics mapping $\nu = E/h$. These conventions give different n values for the same physical scale. For the cosmic timeline above, the thermal mapping is appropriate since we are tracking temperature epochs. For the ring geometry and antipodal structure, the particle physics mapping is used. Both are internally consistent within their respective contexts.

2.1 The Equilibrium Was Crossed Once

The C = 0 equilibrium point ($n_{\text{flip}} = 3.561$, confirmed by 264 BBH events) sits 3.561 octaves from the Big Bang. In physical terms:

$n_{\text{flip}} = 3.561$ corresponds to:

$$\nu_{\text{flip}} = \nu_{\text{P}} / 2^{3.561} = 2.952 \times 10^{42} / 11.80 = 2.502 \times 10^{41} \text{ Hz}$$

$$E_{\text{flip}} = h\nu_{\text{flip}} = 1.034 \times 10^{18} \text{ GeV (just below Planck energy)}$$

$$T_{\text{flip}} = h\nu_{\text{flip}}/k_B = 1.20 \times 10^{31} \text{ K}$$

The equilibrium $C = 0$ corresponds to $T \approx 1.20 \times 10^{31} \text{ K}$.

The Planck temperature is $1.417 \times 10^{32} \text{ K}$.

n_{flip} is 3.561 octaves below the Planck wall:

$$T_{\text{Planck}} / T_{\text{flip}} = 2^{3.561} = 11.80$$

$$T_{\text{flip}} = 1.417 \times 10^{32} / 11.80 = 1.20 \times 10^{31} \text{ K}$$

This is the temperature at which the universe crossed $C = 0$.

It happened in the first $\sim 10^{-41}$ seconds after the Big Bang.

The universe passed through its own equilibrium almost instantly — and has been moving away from it ever since.

2.2 The Robertson Arc

The full traversal of the ring is a Robertson arc: the universe moves from the Robertson maximum ($C = +0.5$ at the Big Bang) through the equilibrium ($C = 0$ at $n = 3.561$) and continues outward to the Robertson maximum again ($C = +0.5$ at the Hubble horizon). It never returns to equilibrium. It never crosses $C = 0$ again after that first instant.

Robertson arc (physical compactness C , not C_{eff}):

Big Bang:	$C = +0.5$ (Planck: every particle inside its Schwarzschild radius)
$n = 3.561$:	$C = 0.0$ (equilibrium, crossed once, $T \approx 1.20 \times 10^{31} \text{ K}$)
$n = 5.314$:	$C = +0.5$ (first Robertson maximum in n -space, BBH scale)
$n = 100.935$:	$C = ?$ (cavity center, matter-forming epoch)
$n = 201.87$:	$C = +0.5$ (Hubble horizon, present-day Robertson maximum)

The arc goes: $+0.5 \rightarrow 0 \rightarrow$ (outward through all of physics) $\rightarrow +0.5$

The universe brackets its entire history between two Robertson maxima with one equilibrium crossing near the very beginning.

3. Three Robertson Maxima on the Ring

[DERIVED]

DERIVED (Big Bang + Hubble) | CONFIRMED (n_{BBH})

The ring contains three distinct Robertson maxima — three points where $C = +0.5$ in physical compactness terms:

Robertson Maximum	n	Physical Identity	Status
Planck wall (Big Bang)	0.00	Every Planck-epoch particle inside its Schwarzschild radius	DERIVED
n_BBH (BH merger scale)	5.314	Black hole horizon compactness; LQG Robertson minimum	CONFIRMED (264 events)
Hubble horizon (present)	201.87	Observable universe IS its own Schwarzschild radius	DERIVED

These three maxima are not equivalent in origin. The Big Bang maximum is a consequence of extreme energy density at the Planck scale (every particle is a micro-black-hole). The BBH maximum is the LQG Robertson minimum for spin-network patches ($j = 1/2$). The Hubble maximum is a consequence of critical density in a flat universe. Three different physical mechanisms. One compactness value: $C = +0.5$.

3.1 Why $C = +0.5$ Appears Three Times

The Robertson minimum $\Delta x \cdot \Delta p \geq \hbar/2$ applied to a gravitational system of mass M gives the minimum radius $r_s = \hbar/(2Mc)$. Setting $GM/rc^2 = \text{compactness}$:

At the Robertson minimum $r = \ell_P$ (Planck scale, Big Bang):
 $C = GM/(r_P c^2)$ with $M = m_P/2$ (from Robertson: $4GM^2 = \hbar c$)
 $C = G(m_P/2)/(\ell_P c^2) = 1/2 = +0.5 \quad \checkmark$

At the LQG $j=1/2$ minimum (BBH scale):
The Seraphim derivation: $n_{\text{BBH}} = 5.314$
 $C_{\text{component}} = 0.5$ exactly (derived in Paper 2 Sec 7.1a) $\checkmark$

At the Hubble horizon (present day):
 $C_H = 1/2$ (from critical density, Zero Point paper Sec 4.1) $\checkmark$

All three are the same Robertson condition:
The minimum position uncertainty = Schwarzschild radius
 $\Delta x_{\text{min}} = r_s \iff C = +0.5$

$C = +0.5$ IS the Robertson minimum expressed as compactness.

Core result:

C = +0.5 is not a boundary condition imported from general relativity. It is the Robertson minimum uncertainty principle expressed as compactness. Every time C = +0.5 appears on the ring — at the Big Bang, at black holes, at the Hubble horizon — it is the same statement: the system's position uncertainty equals its Schwarzschild radius. The universe reaches C = +0.5 at three scales because quantum mechanics demands C = +0.5 wherever a gravitational system reaches its minimum measurable size.

4. The Equilibrium Crossing: C = 0 at the Dawn of Physics

[DERIVED]

DERIVED from confirmed n_{flip} + timeline mapping

$n_{\text{flip}} = 3.561$ is the most confirmed result in the framework: 264 BBH events, 0.05σ , zero free parameters. It is the C = 0 equilibrium. Mapping it onto the cosmic timeline gives:

$n_{\text{flip}} = 3.561$ octaves above the Planck wall

Temperature: $T_{\text{flip}} = T_{\text{Planck}} / 2^{3.561} = 1.417 \times 10^{32} / 11.80 = 1.20 \times 10^{31} \text{ K}$

Age of universe at this temperature:

In radiation-dominated era: $t \propto 1/T^2$

$$\begin{aligned} t_{\text{flip}} &\approx t_{\text{Planck}} \times (T_{\text{Planck}}/T_{\text{flip}})^2 \\ &= 5.39 \times 10^{-44} \text{ s} \times (1.417 \times 10^{32} / 1.20 \times 10^{31})^2 \\ &= 5.39 \times 10^{-44} \times (11.81)^2 \\ &= 5.39 \times 10^{-44} \times 139.5 \\ &\approx 7.52 \times 10^{-42} \text{ seconds} \end{aligned}$$

The universe crossed C = 0 at $t \approx 7.5 \times 10^{-42} \text{ s}$ ($\sim 10^{-41} \text{ s}$).

This is ~ 130 Planck times after the Big Bang.

The equilibrium was crossed in the first blink of existence.

[CORRECTED: original stated 10^{-39} s due to a double-squaring error: the ratio $(T_{\text{Planck}}/T_{\text{flip}})^2 = 11.81^2 = 139.5$ was mistakenly re-squared as $141.2^2 = 1.99 \times 10^4$, inflating t_{flip} by a factor of ~ 142 . Correct result: $\sim 7.5 \times 10^{-42} \text{ s}$.]

4.1 What Happened at the Equilibrium Crossing

At $n = n_{\text{flip}} = 3.561$, the compactness transitions from positive ($C > 0$, gravitational collapse geometry) to the realm where the LQG framework defines the boundary between the two geometric sectors. The Seraphim framework identifies n_{flip} as the 'realm boundary' — the universal lower bound below which no BBH merger can occur.

In cosmic terms, the equilibrium crossing at $T \approx 1.20 \times 10^{31}$ K marks the moment when the universe first became geometrically distinct from its own Planck-scale initial condition. Before n_{flip} : every scale was inside its own Schwarzschild radius ($C \geq +0.5$). At n_{flip} : the first scale emerged where $C = 0$ — where geometry was balanced between collapse and expansion. This is the origin of the distinction between gravity and expansion that defines all subsequent cosmic evolution.

4.2 The Universe Has Never Returned to $C = 0$

From $n_{\text{flip}} = 3.561$ onward, the octave coordinate increases monotonically with cosmic time. $C_{\text{eff}} = (n - 3.561)/3.506$ increases throughout the history of the universe. The $C = 0$ crossing was a one-time event. The universe has been moving away from equilibrium — toward larger n , lower temperature, lower compactness — ever since.

C_{eff} at key epochs (framework compactness coordinate):

Big Bang ($n=0$):	$C_{\text{eff}} = -1.016$ (vacuum mode)
Equilibrium crossing ($n=3.561$):	$C_{\text{eff}} = 0.000$ (crossed once, $T \approx 1.20 \times 10^{31}$ K)
BBH scale ($n=5.314$):	$C_{\text{eff}} = +0.500$ (first maximum)
Electroweak ($n=56.76$):	$C_{\text{eff}} = +15.17$
QCD/proton era ($n=66.14$):	$C_{\text{eff}} = +17.85$
H condensation ($n=92.19$):	$C_{\text{eff}} = +25.28$
CMB ($n=105.36$):	$C_{\text{eff}} = +29.04$
Today / Hubble ($n=201.87$):	$C_{\text{eff}} = +56.56$

C_{eff} only increases. The equilibrium is in the past.

The Hubble horizon is the current C_{eff} maximum.

The universe approaches $C = +0.5$ (physical) again at the horizon.

5. The Möbius Fold: What Happens at the Hubble Horizon

[SPECULATIVE]

SPECULATIVE

The Big Bang and Hubble horizon are both $C = +0.5$ boundaries. On a simple closed ring this would mean the universe loops back to its own beginning. But the ring is not closed in the sense of being periodic — the $N/\Delta n$ ratio is irrational (quasicrystal paper). Something else is happening at the boundary.

5.1 The Fold, Not the Loop

The Möbius topology hypothesis (Convention Bridge paper) proposes that the Hubble horizon is a fold point, not a loop point. The ring doesn't connect the Big Bang back to the Big Bang — it folds. The result of the fold is not a return to $n = 0$ but a forced return to $C = 0$. The Robertson

maximum at the Hubble horizon cannot be sustained — the only available geometric move from $C = +0.5$ (the maximum) with no further expansion possible is return toward the equilibrium.

The physical interpretation: the Hubble horizon is not where the universe ends. It is where the Robertson boundary forces a geometric decision. Beyond $C = +0.5$, physics as described by the LQG framework is undefined (you're inside a Schwarzschild radius). The only defined continuation is back toward $C = 0$.

5.2 The Test: Gaps Do Not Fall at Fold Quarter-Points

A clean prediction of the Möbius fold hypothesis is that geometric gaps (forbidden zones, quasicrystal voids) should cluster near the fold quarter-points ($n = 100.935$ and $n = 0/201.87$). The numerical test found no such clustering:

Gap / Forbidden Zone	n	Distance to QP1 (100.935)	Assessment
GUT desert lower wall	9.581	−91.4 octaves	Far from fold geometry
GUT desert center	14.561	−86.4 octaves	Far from fold geometry
GUT desert upper wall	19.558	−81.4 octaves	Far from fold geometry
Alpha antipode gap	64.780	−36.2 octaves	Between QP0 and QP1, no alignment
H condensation ceiling	92.190	−8.7 octaves	Near QP1 but within matter cluster
Cavity center	100.935	0.0 octaves	IS the quarter-point

Gaps are not fold artifacts. They are quasicrystal forbidden zones (derived in the Quasicrystal paper) determined by the two-phase grid structure, not by the Möbius fold geometry. The fold hypothesis explains the Robertson maximum identification but does not explain gap positions. Gap positions are already explained by the quasicrystal derivation.

6. The Complete Picture: One Ring, One History, One Principle

The universe is a single Robertson arc:

$C = +0.5$ (Big Bang) $\rightarrow C = 0$ (equilibrium, $t \approx 7.5 \times 10^{-42}$ s) $\rightarrow$ [all of physics] $\rightarrow C = +0.5$ (Hubble)

The ring is not a mathematical tool for organizing data. The ring IS the universe's path through compactness space, from its own Robertson maximum at the beginning, through the one moment of equilibrium, to its Robertson maximum at the horizon. Every physical structure — particles, atoms, stars, galaxies, gravitational waves — is a node on this path. The Seraphim octave coordinate n is a universal clock: it measures how far the universe has traveled from its own beginning along the Robertson arc.

Result	Status	Source
$C_{\text{Hubble}} = +0.5$ (Hubble is Robertson maximum)	DERIVED	Critical density + Robertson
$C_{\text{BigBang}} = +0.5$ (Planck epoch is Robertson maximum)	DERIVED	Every particle inside own Schwarzschild radius
Big Bang and Hubble are same $C=+0.5$ boundary	DERIVED	Both follow from Robertson minimum
$n_{\text{flip}} = 3.561$ is $C = 0$ equilibrium	CONFIRMED	264 BBH events, 0.05σ
Equilibrium crossed at $T \approx 1.20 \times 10^{31}$ K, $t \approx 7.5 \times 10^{-42}$ s	DERIVED	$n_{\text{flip}} \rightarrow T$ via frequency mapping
Ring traversal = cosmic timeline $n = 0$ to 201.87	DERIVED	Frequency-temperature mapping of all epochs
Gaps are quasicrystal forbidden zones, not fold artifacts	DERIVED	Numerical test: no fold quarter-point alignment
Möbius fold at Hubble forces return to $C = 0$	SPECULATIVE	Consistent geometry; not formally derived
Universe has never returned to $C = 0$ after $t \approx 7.5 \times 10^{-42}$ s	DERIVED	C_{eff} increases monotonically with n

7. Priority Statement

Companion to the Seraphim LQG Framework (DOI: 10.5281/zenodo.18852768). March 28, 2026.

Priority claims (March 28, 2026):

- The identification of the Big Bang boundary (Planck wall, $n = 0$) and the Hubble horizon ($n = 201.87$) as the same $C = +0.5$ Robertson maximum, approached from opposite ends of the Seraphim ring.
- The full cosmic timeline mapped to the Seraphim octave coordinate: Planck epoch ($n \approx 0.5$) through GUT ($n \approx 13.6$), electroweak ($n \approx 56.8$), QCD/proton era ($n \approx 66.1$), BBN ($n \approx 76.9$), recombination ($n \approx 95.3$), CMB ($n \approx 105.4$), to the Hubble horizon ($n = 201.87$).
- The derivation that the $C = 0$ equilibrium ($n_{\text{flip}} = 3.561$, confirmed) corresponds to $T \approx 1.20 \times 10^{31}$ K, crossed at $t \approx 7.5 \times 10^{-42}$ seconds after the Big Bang — the one moment the universe passed through its own equilibrium.
- The three independent Robertson maxima on the ring: Planck/Big Bang (energy density), n_{BBH} (LQG area minimum), and Hubble horizon (critical density). Three mechanisms, one compactness: $C = +0.5 =$ Robertson minimum expressed as compactness.
- The numerical confirmation that gaps (GUT desert, alpha antipode) do not align with Möbius fold quarter-points — they are quasicrystal forbidden zones, not fold artifacts.

Correction log (corrected reprint):

- (1) Abstract and Section 2.1: $T \approx 4 \times 10^{\text{(corrupted)}}$ corrected to $T \approx 1.20 \times 10^{31}$ K.
- (2) Section 4 box: t_{Planck} exponent corrected from corrupted 10^7 to 10^{-44} ; double-squaring error fixed: $(T_{\text{P}}/T_{\text{flip}})^2 = 11.81^2 = 139.5$, not 141.2^2 ; t_{flip} corrected from 1.07×10^{-39} s to 7.5×10^{-42} s ($\sim 10^{-41}$ s). Propagated to Section 2.1, Section 4.1 label, summary table, and Priority Statement.
- (3) Section 1.2: $\phi_1 = 0.055$ description corrected — deviation of $C_{\text{eff}}(0)$ from -1.000 is 0.016 ; $0.055 = n_{\text{flip}} - \text{slope}$ is a framework constant coincidence, not the numerical deviation. Text rewritten to distinguish the two.
- (4) Timeline QCD row: added convention note — $n = 66.14$ uses thermal mapping (kT); ring landmark $n_{\text{QCD}} = 68.800$ uses particle physics mapping ($E = h\nu$). No framework constants, prediction values, or confirmed results were changed.

End of Derivation Paper | The Same Boundary | March 28, 2026 | v1 (corrected reprint)

"The universe began at its Robertson maximum. It crossed equilibrium once, $\sim 10^{-41}$ seconds in. It has been moving away from $C = 0$ ever since. It will reach its Robertson maximum again at the Hubble horizon. The ring is the journey between."